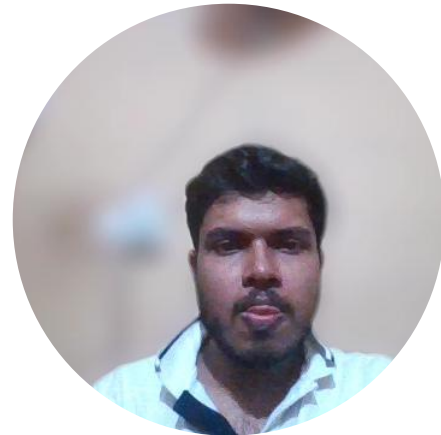


Medibox Enhancement Light Sensitivity and Control

Building on Success

210494D

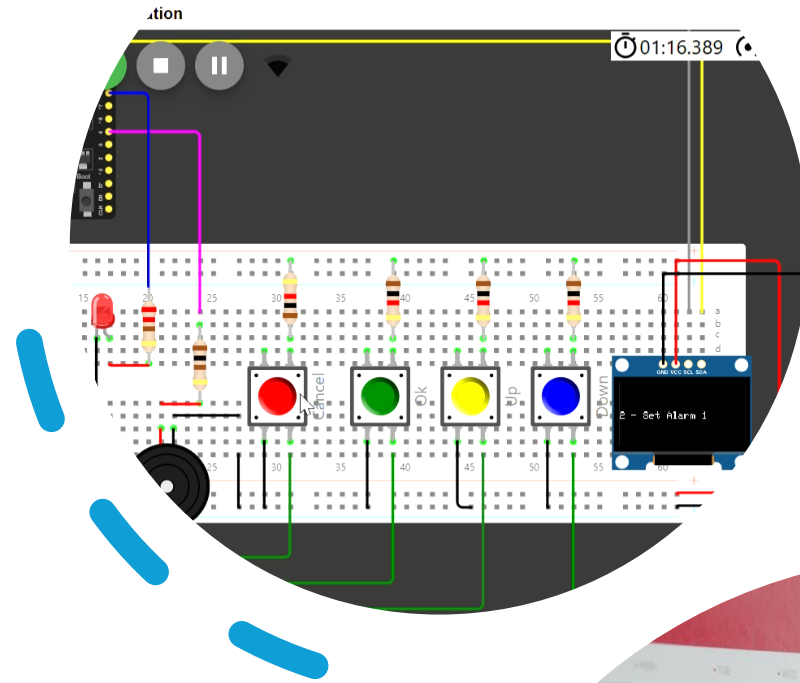
PREMAKUMARA HPS



Recap of Part 1

Key Achievements:

- ✓ Successfully simulated Medibox core functions on Wokwi
- ✓ Implemented time zone setting, alarms, NTP time sync, and basic health monitoring



WOKWI

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210494D_PREMAKUMARA HPS_MEDIBOX_PART1

Docs

Main_program.ino

diagram.json

libraries.txt

Library Manager

```
1 //include libraries
2 #include <DHTesp.h>
3 #include <Wire.h>
4 #include <Adafruit_GFX.h>
5 #include <Adafruit_SSD1306.h>
6 #include <WiFi.h>
7 #include <sys/time.h>
8
9 //declare OLED parameter
10 #define screen_width 128
11 #define screen_height 64
12 #define OLED_reset -1
13 #define screen_address 0x3C
14
15 //declare wifi time parameters
16 const char* ssid = "Wokwi-GUEST";
17 const char* password = "";
18 int wifi_channel=6;
19
20 //define for healthy condition for medicine
21 #define TEMP_HIGH 32
22 #define TEMP_LOW 26
23 #define HUMIDITY_HIGH 80
24 #define HUMIDITY_LOW 60
25
26 //declare buzzer parameters
27 int notes[] = {262, 294, 330, 349, 392, 440, 494, 523}; //C,D,E,F,G,A no 0 call 12 Room 4
28 int n_notes = sizeof(notes) / sizeof(notes[0]);
29
30 //declare pins
31 #define DHT_PIN 14
32 #define LED_PIN 5
```

Simulation

01:00.088 84%

load:0x40080400,len:2972
entry 0x400805dc



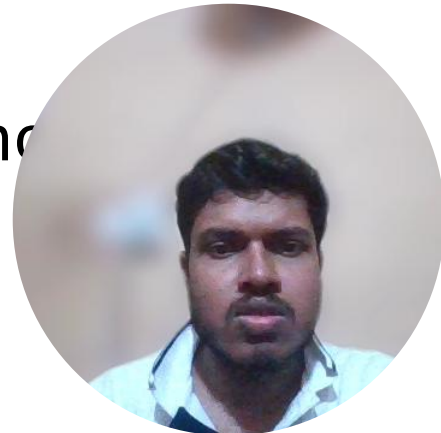
Why need enhancement ?

Problem:

- Certain medications are sensitive to light exposure
- Inconsistent light levels can affect medication efficacy and safety

Solution:

- Implement a smart light monitoring and control system for the Medibox
- Use LDRs for sensing, a servo motor for control, and Node-RED for user interface and data visualization

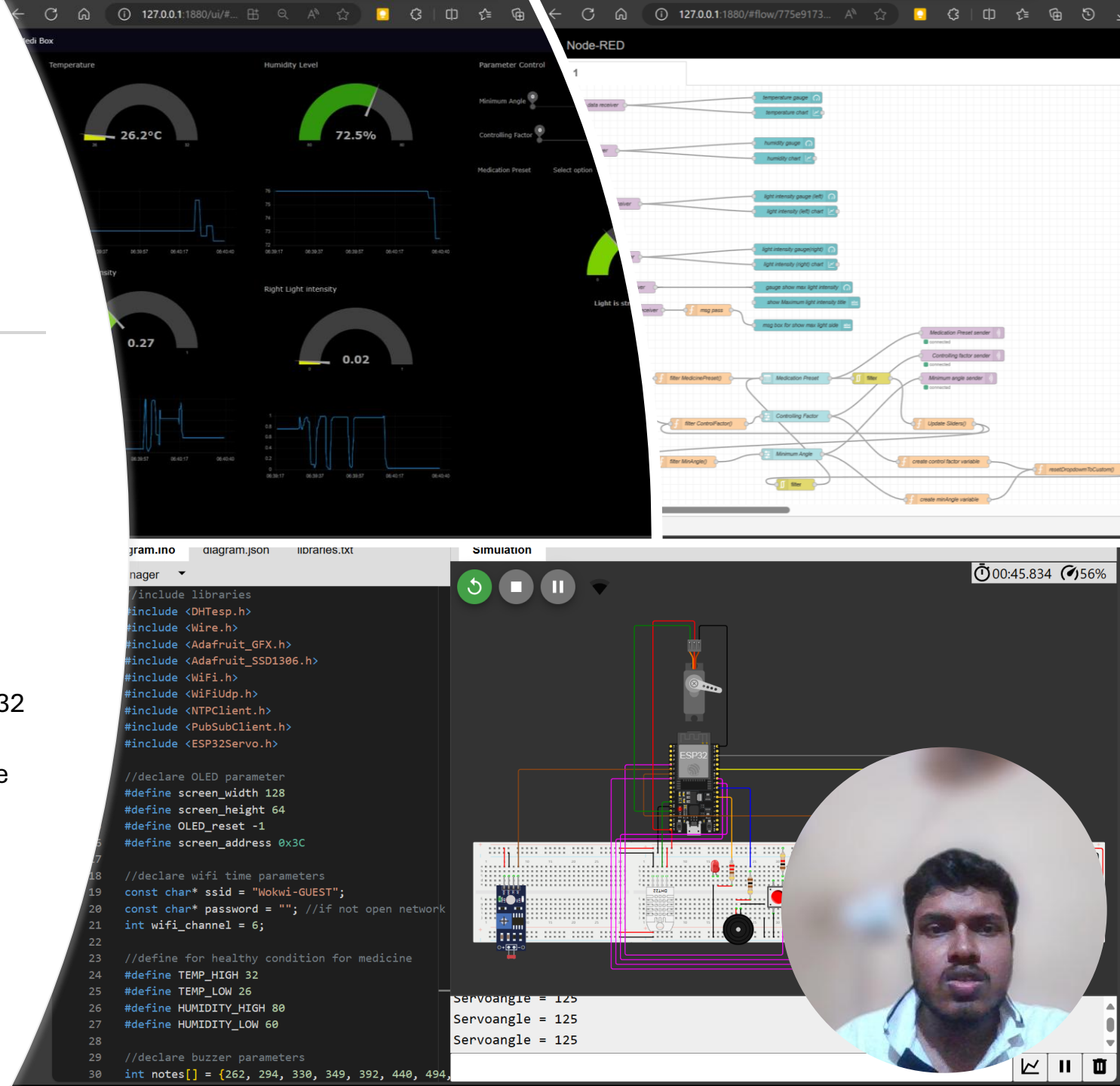


System Overview

- ✓ The Medibox with two LDRs (left and right)
- ✓ The servo motor controlling a shaded window
- ✓ The ESP32 connected to the LDRs and servo motor
- ✓ The Node-RED dashboard interface

Key Components:

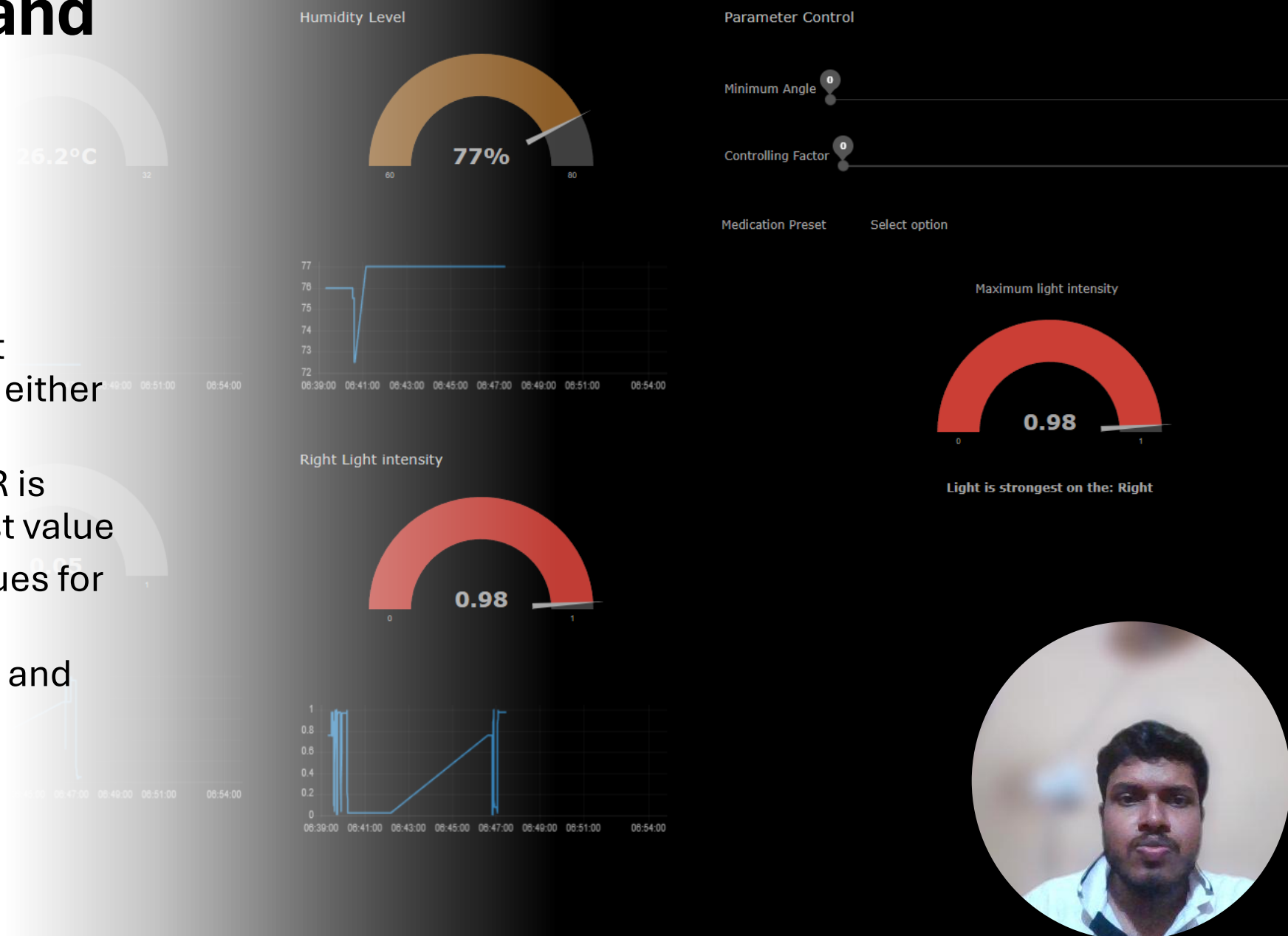
- ❑ ESP32: Microcontroller for data acquisition and control
- ❑ Wokwi: Simulation environment for testing the ESP32 code
- ❑ Node-RED: Visual programming tool for creating the dashboard
- ❑ LDRs: Light sensors
- ❑ Servo motor: Controls the window position



Light Sensing and Visualization

Key Features:

- ❑ Displays the highest intensity value from either LDR
- ❑ Indicates which LDR is providing the highest value
- ❑ Scales intensity values for clear visualization
- ❑ Scales Temperature and humidity level also



```

#####23
void updateLight()
{
    int rawLightLeft = analogRead(LDR_LEFT_PIN); // Read raw LDR sensor values
    int rawLightRight = analogRead(LDR_RIGHT_PIN);

    // Normalize light values to 0-1 range (with calibration)
    float normalizedLightLeft = (rawLightLeft - LDR_MAX) / (LDR_MIN - LDR_MAX);
    float normalizedLightRight = (rawLightRight - LDR_MAX) / (LDR_MIN - LDR_MAX);

    // Determine highest light intensity and its source
    float highestLightIntensity = max(normalizedLightLeft, normalizedLightRight);
    String lightSource = (normalizedLightLeft > normalizedLightRight) ? "Left" : "Right";

    // Calculate servo angle based on light conditions
    float directionFactor = (lightSource == "Right") ? 0.5 : 1.5;
    int servoAngle = theta_offset * directionFactor + (180 - theta_offset) * highestLightIntensity * controllingFactor;
    servoAngle = min(servoAngle, 180); // Limit to 180 degrees
    Serial.print(("Servoangle = "));
    Serial.println(servoAngle);

    // Adjust servo position
    servo.write(servoAngle);
}

```

```

// Adjust servo position
servo.write(servoAngle);

// publish MQTT ldr messages
String(normalizedLightLeft, 2).toCharArray(ldrLeftArr, 6);
mqttClient.publish("210494D/medibox/light_intensity/left", ldrLeftArr );

String(normalizedLightRight, 2).toCharArray(ldrRightArr, 6);
mqttClient.publish("210494D/medibox/light_intensity/right", ldrRightArr );

String(highestLightIntensity, 2).toCharArray(ldrMaxArr, 6);
mqttClient.publish("210494D/medibox/light_intensity/max", ldrMaxArr );

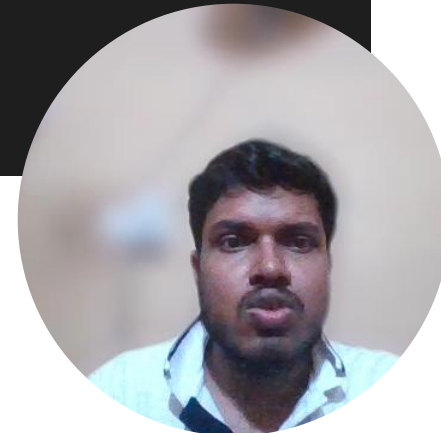
lightSource.toCharArray(ldrMaxSource, 6);
mqttClient.publish("210494D/medibox/light_intensity/max/source", ldrMaxSource );

```

```

683
684 void loop()
685 {
686     if (!mqttClient.connected())
687     {
688         Serial.println("Reconnecting to MQTT Broker");
689         connectToBroker();
690     }
691
692     mqttClient.loop(); //enter mqtt
693
694     update_time_with_check_alarm_and_check_warning(); //###9
695
696     updateLight();
697
698     if (digitalRead(PB_ok) == LOW)
699     {
700         delay(200);
701         go_to_menu(); //###17
702     }
703 }
704

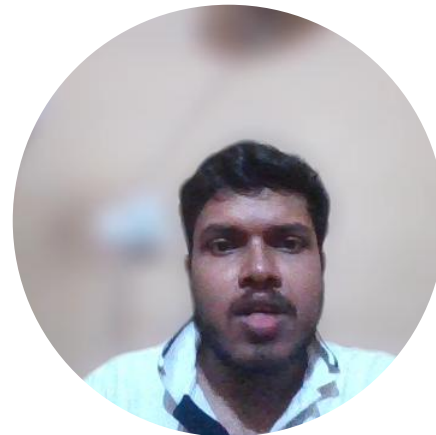
```



```
void connectToBroker()
{
    while (!mqttClient.connected())
    {
        Serial.println("Attempting MQTT connection... ");
        if (mqttClient.connect("ESP32Client-465456464S645"))
        {
            Serial.println("Connected");
            mqttClient.subscribe("210494D/nodeRed/angle/min"); //subscribe
            mqttClient.subscribe("210494D/nodeRed/controlling_factor");
            mqttClient.subscribe("210494D/nodeRed/medication_preset");
        }

        else
        {
            Serial.println("Failed to connect, rc=");
            Serial.print(mqttClient.state());
            delay(5000);
        }
    }
}
```

```
//###21
void setupMqtt()
{
    mqttClient.setServer("test.mosquitto.org", 1883);
    mqttClient.setCallback(receiveCallback);
}
```



```

#####20
void receiveCallback(char* topic, byte* payload, unsigned
{
    Serial.print("Message arrived [");
    Serial.print(topic);
    Serial.print("] ");

    char payloadCharArray[length];

    for (int i = 0; i < length; i++)
    {
        Serial.print((char)payload[i]);
        payloadCharArray[i] = (char)payload[i];
    }

    Serial.println();

    int temp_theta_offset = 0;
    float temp_controllingFactor = 0;

    if (strcmp(topic, "210494D/nodeRed/angle/min") == 0)
    {
        temp_theta_offset = String(payloadCharArray).toInt();
        Serial.println(temp_theta_offset);
    }
}

```

```

int temp_theta_offset = 0;
float temp_controllingFactor = 0;

if (strcmp(topic, "210494D/nodeRed/angle/min") == 0)
{
    temp_theta_offset = String(payloadCharArray).toInt();
    Serial.println(temp_theta_offset);
}

else if (strcmp(topic, "210494D/nodeRed/controlling_factor") == 0)
{
    temp_controllingFactor = String(payloadCharArray).toFloat();
    Serial.println(temp_controllingFactor);
}

else if (strcmp(topic, "210494D/nodeRed/medication_preset") == 0)
{
    String preset = String(payloadCharArray);

    if (preset == "custom")
    {
        //replace with dashboard values
        if (temp_controllingFactor != 0)
        {
            controllingFactor = temp_controllingFactor;
        }

        if (temp_theta_offset != 0)
        {
            theta_offset = temp_theta_offset;
        }
    }
}

```

```

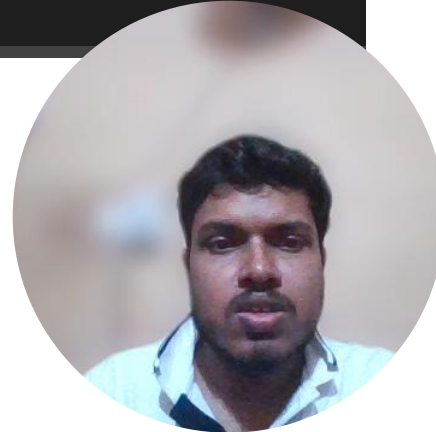
{
    if (temp_theta_offset != 0)
    {
        theta_offset = temp_theta_offset;
    }
}

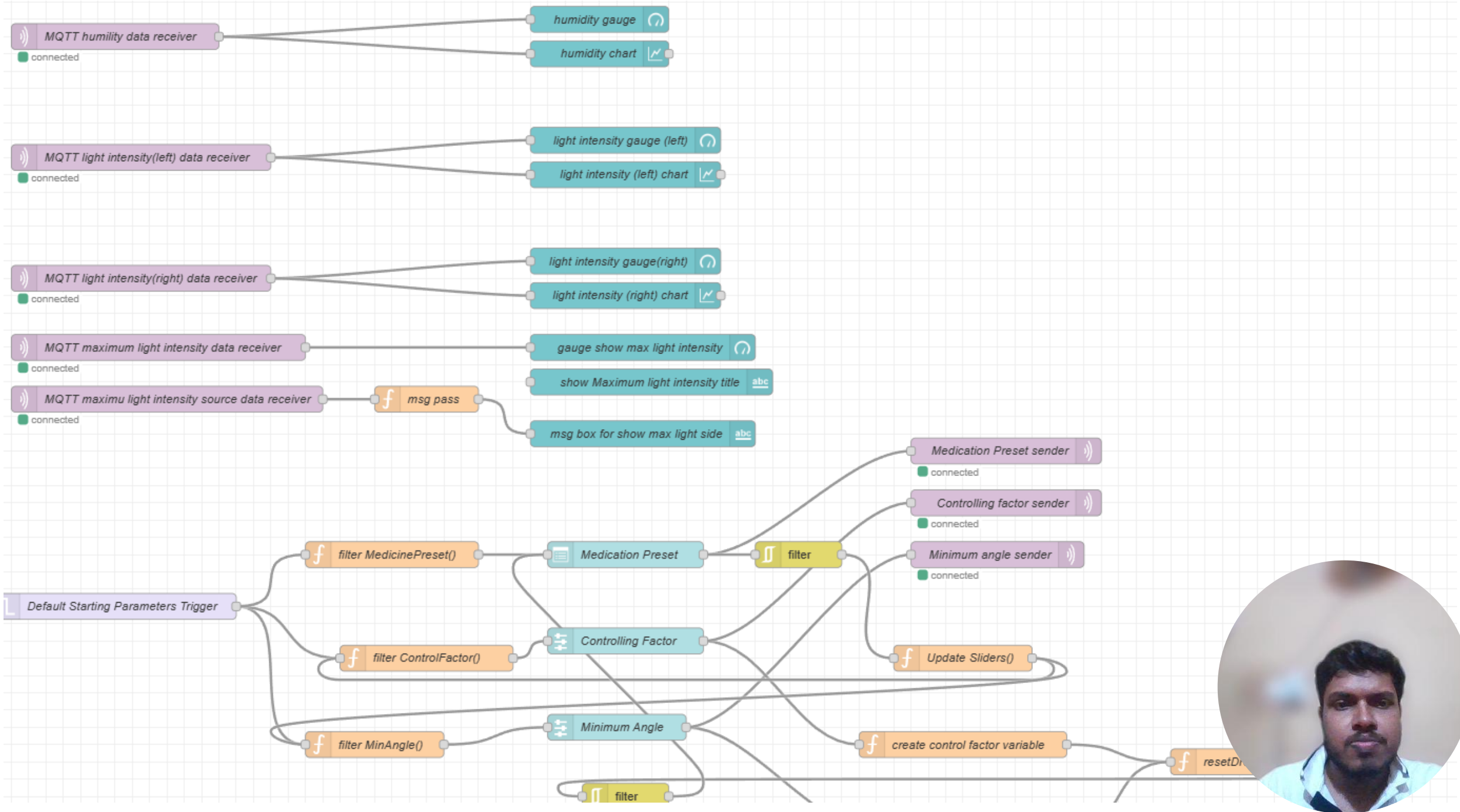
else
{
    // Apply preset values based on the selected medication
    if (preset == "tablet_a")
    {
        theta_offset = 45; // Replace with actual values for Tablet A
        controllingFactor = 0.6;
    }

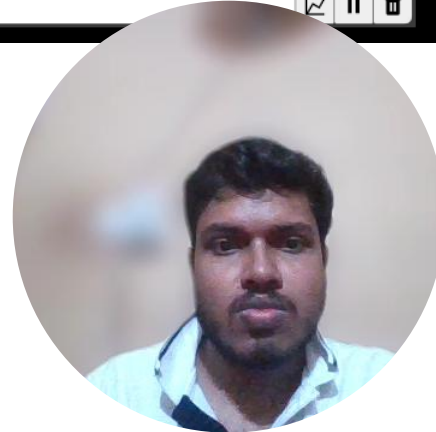
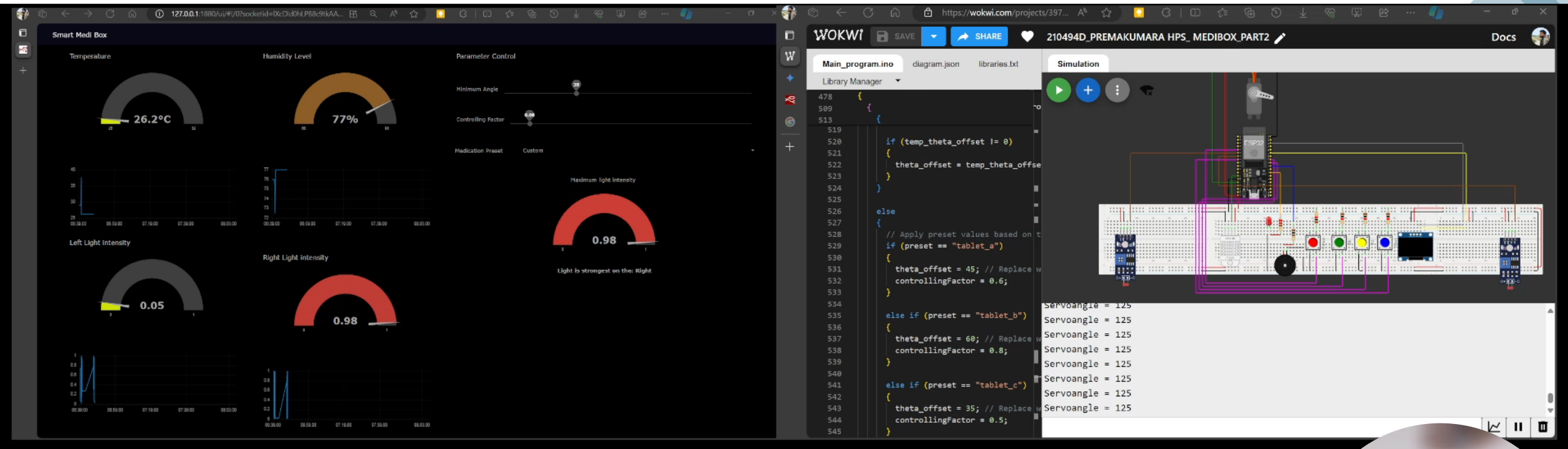
    else if (preset == "tablet_b")
    {
        theta_offset = 60; // Replace with actual values for Tablet B
        controllingFactor = 0.8;
    }

    else if (preset == "tablet_c")
    {
        theta_offset = 35; // Replace with actual values for Tablet C
        controllingFactor = 0.5;
    }
}
}

```







Smart Medi Box

Temperature

29.7°C

Left Light Intensity

0.67

Humidity Level

68%

Right Light Intensity

0.76

Parameter Control

Minimum Angle

Controlling Factor

Medication Preset

Custom

Maximum light Intensity

0.76

Light is strongest on the Right

WOKWI

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210494D_PREMAKUMARA HPS_MEDIBOX_PART2

Docs

Main_program.ino

diagram.json

libraries.txt

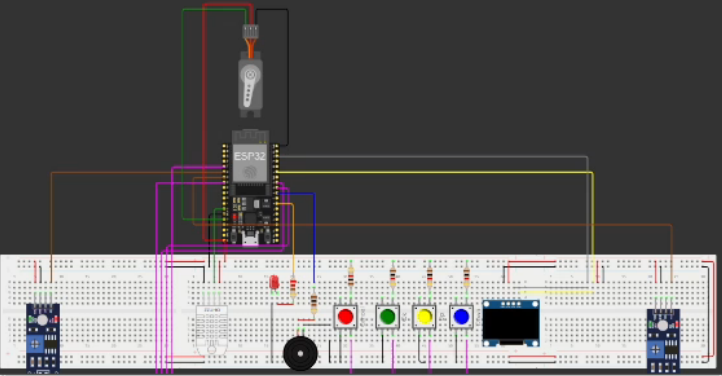
Library Manager

```
118 {
137 {
139   Serial.println(temp_theta_offset);
140 }
141
142 else if (strcmp(topic, "210494D/nodeR
143 {
144   temp_controllingFactor = String(pay
145   Serial.println(temp_controllingFact
146 }
147
148 else if (strcmp(topic, "210494D/nodeR
149 {
150   String preset = String(payloadCharA
151
152   if (preset == "custom")
153   {
154     //replace with dashboard values
155     if (temp_controllingFactor != 0)
156     {
157       controllingFactor = temp_contro
158     }
159
160     if (temp_theta_offset != 0)
161     {
162       theta_offset = temp_theta_offse
163     }
164   }
165
166   else
```

Simulation

00:10.830

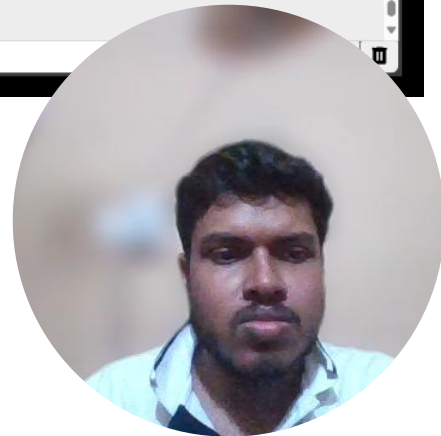
19%



Servoangle = 100

Servoangle = 100

Servoangle = 100



Thank you

